# Do Now · retrieval

## Dijkstra: shortest paths I

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

Why does quicksort exclude the pivot from its recursive calls?

### Q2

Define a weighted edge.

### Q3

What is the first index of an array?

### Q4

A–B costs 3 and B–C costs 5. What is cost of A→B→C?

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Initialise from the start — Distance(A)=0, all others infinity. Relax A's edges to set B=4 and C=2.

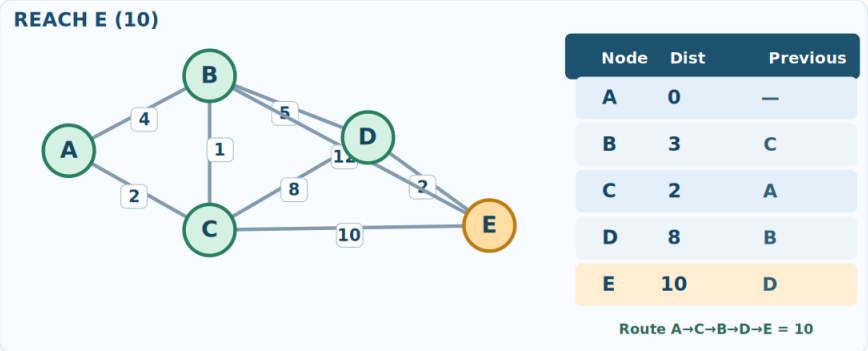
## Core idea 2

Always choose smallest tentative distance — C=2 settles before B=4. Through C, B improves to 3.

## Core idea 3

Keep updating neighbours — Settling B changes D from 10 to 8; settling D changes E from 12 to 10.

**Key visual · complete concept**



# New Learning · applying the ideas

## Core idea 4

Reconstruct from predecessors — E←D←B←C←A. Reverse these arrows for A→C→B→D→E, total distance 10.

## Core idea 5

Dijkstra's requirement — Edge weights must be non-negative. A visited/settled node's shortest distance is then final.

## Vocabulary · use the correct technical terms

tentative distance — Best known cost to an unsettled vertex.

settled — Vertex whose shortest distance has been finalised.

relaxation — Improve a neighbour's distance via the current vertex.

predecessor — Previous vertex on the best-known route.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

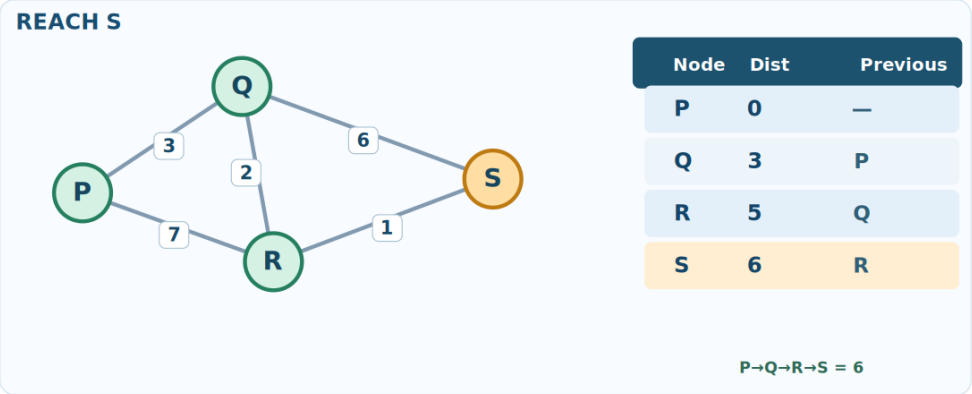
## Problem

Find the shortest P→S route using non-negative weights.

## Follow the worked method

1. Predict the next state — P gives tentative Q=3 and R=7.
2. Reveal and explain — Settle R=5, improving S to 6.
3. Reveal and explain — Predecessors S←R←Q←P give P→Q→R→S, cost 6.

**Source model · final illustrated step**




| Node | Dist | Previous |
|---|---|---|
| P | 0 | — |
| Q | 3 | P |
| R | 5 | Q |
| S | 6 | R |

## Why this method works

Identify the decision, rule or invariant that makes the method correct.

# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict the next state
From M, set N=5 and O=2; O is the next vertex settled.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

# You Do · independent practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

## Question 1

A–B=4 and A–C=2. Starting Dijkstra at A, state the tentative distances of A, B and C after relaxing A's neighbours.

## Question 2

On edges S–A=3, S–B=7, A–B=2, B–T=2 and A–T=8, find the shortest S→T path with full distance reasoning.

Check that you have used the technical terms and explained your decisions.

# You Do · extended practice

Answer every question in BLACK PEN. Show steps and use OCR command words.

## Question 3

Explain why Dijkstra must not be relied upon when negative-weight edges are possible, and contrast tentative and settled distances.

Check that you have used the technical terms and explained your decisions.

# Exit Ticket · reflect and retrieve

## Exit Ticket

Dijkstra always chooses the next vertex by its smallest edge weight.

## Use key vocabulary from memory

1. Explain tentative distance in your own words.

2. Explain settled in your own words.

3. Explain relaxation in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.